

Titanic Dataset – Exploratory Data Analysis (EDA) Report

1. Introduction

This report presents an exploratory data analysis of the Titanic dataset. The purpose is to understand patterns, trends, and relationships among variables that influenced passenger survival.

2. Dataset Overview

The dataset contains demographic and travel information for Titanic passengers, including Age, Sex, Fare, Embarked, Pclass, and Survival status.

3. Key Statistics

- Age ranges mostly between 20–40 years.
- Majority of passengers belonged to 3rd class.
- More males onboard than females.
- Fare distribution is right-skewed.

4. Univariate Analysis – Summary

- Age: Right-skewed distribution; majority are young adults.
- Fare: Highly right-skewed; few passengers paid premium fares.
- Sex: More males than females.
- Survived: Dataset is imbalanced with more non-survivors.

5. Bivariate Analysis – Summary

- Females had far higher survival rates than males.
- Higher-class passengers (Pclass 1) had better survival.
- Higher fares correlate with greater survival.
- Younger passengers (children) had slightly higher chance of survival.

6. Multivariate Insights

- Strong negative correlation between Fare and Pclass.

- Survival correlates positively with Fare and female gender.
- Age shows weak correlation with survival.

7. Missing Values

- Age: Missing and requires median imputation.
- Cabin: Mostly missing; not useful for modeling.
- Embarked: Two missing values; fill with mode.

8. Final Conclusions

Survival on the Titanic was strongly influenced by:

- Gender (females prioritized during rescue)
- Passenger class (higher class = better chances)
- Fare (correlates with socioeconomic status)

This EDA highlights demographic and socioeconomic factors that played major roles in survival outcomes.